

Stress, Strain & Material Behavior

$\sigma = \frac{P}{A}$ – average normal stress, axial force $P \perp$ to area A ;
 $\tau_{avg} = \frac{V}{A}$ – average shear stress, V parallel (in-plane) to A
 $\varepsilon = \frac{\delta}{L} = \frac{\Delta L}{L}$ – normal strain (elongation per unit length);
 $\gamma = \frac{\pi}{2} - \theta' - \theta$ – shear strain, change in angle between originally $\perp$ lines (rad)

$\varepsilon_T = \alpha \Delta T$ – thermal strain; α = coefficient of thermal expansion ($1/^\circ\text{C}$). Causes no stress unless deformation is restrained

Stress-strain diagram: proportional limit (Hooke's Law valid) $\rightarrow$ elastic limit $\rightarrow$ yield $\sigma_Y \rightarrow$ ultimate σ_u (max) $\rightarrow$ fracture. Ductile: yielding & necking; brittle: fractures near the elastic limit, little plastic deformation

$\sigma = E\varepsilon$, $\tau = G\gamma$ – Hooke's Law (below the proportional limit);
 E = Young's modulus, G = shear modulus

$\nu = -\frac{\varepsilon_{lat}}{\varepsilon_{long}}$ – Poisson's ratio (≈ 0.25 – 0.35 for metals); $G = \frac{E}{2(1+\nu)}$ – only 2 of E, G, ν independent (isotropic)

$$\varepsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)], \quad \varepsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)]$$

$$\varepsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)], \quad \gamma_{xy} = \frac{\tau_{xy}}{G}$$

– generalized Hooke's Law, isotropic (3D); also $\gamma_{yz} = \frac{\tau_{yz}}{G}$, $\gamma_{xz} = \frac{\tau_{xz}}{G}$

$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y)$, $\varepsilon_y = \frac{1}{E}(\sigma_y - \nu\sigma_x)$, $\gamma_{xy} = \frac{\tau_{xy}}{G}$ – generalized Hooke's Law, plane stress ($\sigma_z = \tau_{xz} = \tau_{yz} = 0$)

$\varepsilon_z = -\frac{\nu}{E}(\sigma_x + \sigma_y)$ – plane stress: ε_z is generally NOT zero even though $\sigma_z = 0$

St. Venant's principle: stress concentrations near a load application point or cross-section change die out; the distribution becomes uniform at $\approx$ one cross-sectional dimension away from the discontinuity

Typical elastic constants: $E \approx 200$ GPa, $G \approx 75$ GPa, $\nu \approx 0.32$ – structural steel – $E \approx 70$ GPa, $G \approx 26$ GPa, $\nu \approx 0.35$ – aluminum alloy – $E \approx 114$ GPa, $G \approx 44$ GPa, $\nu \approx 0.31$ – titanium alloy (Ti-6Al-4V), common in aerospace – $E \approx 25$ – 30 GPa, $\nu \approx 0.15$ – 0.2 – concrete (compression)

Axial Deformation

$\delta = \frac{PL}{AE}$ – elongation, axially loaded prismatic member; constant P, A, E over length L

$$\delta = \sum_{i=1}^n \frac{N_i L_i}{A_i E_i}$$

– elongation, multiple segments: sum where N, A , or material is piecewise constant (stepped shafts, composite bars)

$$\delta = \int_0^L \frac{N(x) dx}{A(x) E}$$

– elongation, varying $N(x)$ or $A(x)$: tapered bar, bar under self-weight

$\delta_T = \alpha \Delta T L$ – free thermal expansion/contraction, unrestrained member; combine with $\delta = PL/AE$ by superposition for restrained (indeterminate) members

$\sum F = 0$ (equilibrium), $\delta_{compat} = 0$ or a specified misfit – indeterminate axial members: write equilibrium (fewer eqns than unknowns), then compatibility relating displacements; substitute $\delta = NL/AE$ ($\pm \delta_T$) and solve simultaneously

Misfits/pre-strains: a member fabricated Δ too long/short induces an internal force with zero external load; solve $\delta_{actual} = \delta_{load} \pm \Delta$ by the same compatibility approach

Equilibrium of Beams – Shear & Moment

$\frac{dV}{dx} = -w(x)$, $\frac{dM}{dx} = V(x)$ – load-shear-moment differential relations; $w(x)$ positive downward. Shear slope = –load intensity; moment slope = shear

$V_2 - V_1 = -\int_{x_1}^{x_2} w(x) dx$ – change in shear = –area under the load diagram

$M_2 - M_1 = \int_{x_1}^{x_2} V(x) dx$ – change in moment = area under the shear diagram (fast sketch of V, M without full integration)

Sign convention: positive V tends to rotate a beam element clockwise (up on the left face, down on the right); positive M bends the beam concave up (sagging), tension on the bottom fiber

Procedure: (1) find all support reactions (2) divide the beam into regions between loads/reactions/discontinuities (3) in each region cut a section and solve $V(x), M(x)$ from equilibrium (4) plot V, M vs. x , check slopes against $dV/dx, dM/dx$

$$\langle x-a \rangle^n = \begin{cases} 0, & x < a \\ (x-a)^n, & x \geq a \end{cases} \quad (n \geq 0)$$

– discontinuity (Macaulay) function; brackets "switch on" at $x = a$. Negative-exponent forms ($n = -1, -2$) represent point loads/moments and integrate by raising the exponent by one

Write the entire load $w(x)$ as one expression using Macaulay terms for each load/moment/reaction; integrate once for $V(x) = \int w dx$, again for $M(x) = \int V dx$, term by term – avoids splitting the beam into regions

Common beam cases (quick-check): $R = P/2$ each end, $M_{max} = PL/4$ at midspan – simply supported, central point load P – $R = wL/2$ each end, $M_{max} = wL^2/8$ at midspan – simply supported, uniform load w – $R = P$, $M_{max} = PL$ at the fixed end – cantilever, end point load P – $R = wL$, $M_{max} = wL^2/2$ at the fixed end – cantilever, uniform load w

Stress in Beams – Flexure

$\sigma = -\frac{My}{I}$, $\sigma_{max} = \frac{Mc}{I}$ – flexure formula; y = distance from the neutral axis (through the centroid), c = distance to the extreme fiber. Linear-elastic, symmetric bending

Assumptions: plane sections remain plane; material is linear-elastic (Hooke's Law); neutral axis passes through the centroid; strain/stress vary linearly from zero at the neutral axis

$S = \frac{I}{c}$, $\sigma_{max} = \frac{M}{S}$ – section modulus (purely geometric);

$S_{reqd} = \frac{M_{max}}{\sigma_{allow}}$ – size a section meeting S_{reqd} for beam design

$I = \int y^2 dA$, $I = \bar{I} + Ad^2$ – second moment of area & parallel-axis theorem; $\bar{I}$ about the shape's own centroidal axis, d = perpendicular distance to the parallel axis. Essential for composite/built-up sections

$J = I_x + I_y$, $k = \sqrt{\frac{I}{A}}$ – polar moment of inertia (about the same point, $\perp$ to the x - y plane) & radius of gyration

Second moment of area – common shapes: $\bar{I} = bh^3/12$ – rectangle ($b \times h$), centroidal axis $\parallel$ to base b – $I = bh^3/3$ – rectangle, about the base edge – $\bar{I} = bh^3/36$ – triangle (base b , height h), centroidal axis $\parallel$ base – $I = bh^3/12$ – triangle, about the base edge – $\bar{I} = \pi r^4/4$ – circle (radius r), any centroidal diameter – $J = \pi r^4/2$ – circle, polar, about the center – $I = \pi r^4/8$ – semicircle, about the diametral (straight) edge – $\bar{I} \approx 0.1098 r^4$ – semicircle, centroidal axis $\parallel$ diameter – $I = \pi r^4/16$ – quarter circle, about either straight edge – $\bar{I} \approx 0.0549 r^4$ – quarter circle, centroidal axis $\parallel$ a straight edge

$n = \frac{E_2}{E_1}$, $b' = n b$ – composite/non-homogeneous beams, transformed section: scale material 2's width by n to an equivalent width of material 1, then apply the ordinary flexure formula; multiply the transformed-region stress by n to recover the actual stress there

Quick Reference – at a glance

Elastic constants: $\sigma = E\varepsilon$ – $\tau = G\gamma$ – $G = \frac{E}{2(1+\nu)}$ – $\varepsilon_T = \alpha \Delta T$

Axial & beams: $\delta = \frac{PL}{AE}$ – $\sigma_{max} = \frac{Mc}{I}$ – $S = \frac{I}{c}$ – $\frac{dV}{dx} = -w(x)$ – $\frac{dM}{dx} = V(x)$

Second moment coefficients, centroidal axis: rectangle $bh^3/12$ – triangle $bh^3/36$ – circle $\pi r^4/4$ – semicircle $\approx 0.1098 r^4$ – quarter circle $\approx 0.0549 r^4$

Beam quick-check (M_{max}): simply supported, point load $PL/4$ – simply supported, UDL $wL^2/8$ – cantilever, point load PL – cantilever, UDL $wL^2/2$